Satellite-Based Monitoring of Vegetation Health and Moisture Dynamics

of a Commercial Rice Farm, Nassarawa State, Nigeria

A Multi-Temporal NDVI and NDMI Analysis Using Sentinel-2 Imagery (2023–2025)

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EXECUTIVE SUMMARY

This report presents a multi-temporal satellite-based analysis of vegetation health and canopy moisture dynamics at a commercial rice farm in Nassarawa State, Nigeria, operated by Olam International. The study utilised Sentinel-2 Surface Reflectance imagery acquired over three consecutive growing seasons (2023, 2024, and 2025) to compute the Normalised Difference Vegetation Index (NDVI) and Normalised Difference Moisture Index (NDMI) across a selected farm plot of approximately 61.3 hectares.

Seasonal median composites and full-season time series were generated using Google Earth Engine, enabling both spatial mapping of index distributions and temporal tracking of crop development from planting through harvest. Soil texture analysis was conducted using the iSDAsoil dataset, characterising the farm's surface soils as Sandy Clay Loam with a pH of 6.2.

Key findings indicate variable but generally improving performance across the three seasons. NDVI peaked at 0.691 in 2023, 0.752 in 2024, and 0.773 in 2025, suggesting a progressive trend of increasing canopy density at peak season. NDMI patterns confirmed seasonal moisture dynamics consistent with paddy rice phenology, with negative values at planting rising to positive peaks at canopy closure. Both indices exhibited inter-annual variability that reflects differences in season timing, cloud cover data availability, and likely crop management factors.

The findings demonstrate the value of freely available satellite data for monitoring commercial rice operations at field scale, providing actionable insights on crop performance without requiring ground-based measurements.

1. INTRODUCTION

Rice is one of the most important staple crops in Nigeria, supporting both food security and rural livelihoods across the country. Nigeria is among the largest rice producers in sub-Saharan Africa, yet domestic production continues to fall short of national demand, making the performance optimisation of existing commercial rice operations a priority for agricultural development. Nassarawa State, located in the North-Central geopolitical zone of Nigeria, is an increasingly significant rice-producing region, benefiting from a humid tropical climate, adequate seasonal rainfall, and relatively fertile soils.

Remote sensing has emerged as a powerful and cost-effective tool for monitoring agricultural systems at field and landscape scales. Satellite-derived vegetation indices, particularly the Normalised Difference Vegetation Index (NDVI) and the Normalised Difference Moisture Index (NDMI), provide quantitative measures of crop health and canopy water content respectively, enabling analysts to track crop development, identify stress zones, and assess inter-annual performance variability without requiring direct field access.

The Sentinel-2 satellite constellation, operated by the European Space Agency, provides multispectral imagery at 10-metre spatial resolution with a revisit frequency of approximately five days. This combination of high spatial and temporal resolution makes Sentinel-2 particularly suitable for field-scale agricultural monitoring in tropical regions where cloud cover frequently limits optical data availability.

This study applies a multi-temporal NDVI and NDMI analysis framework to a commercial rice farm in Nassarawa State over three consecutive growing seasons (2023, 2024, and 2025). The objectives of the study are threefold: first, to characterise the spatial distribution of vegetation health and moisture conditions across the farm at seasonal scale; second, to track temporal dynamics of crop development through full-season time series analysis; and third, to assess inter-annual trends in farm performance using satellite-derived evidence.

The methodology, data sources, and analytical workflow are documented in full to ensure reproducibility, and all outputs are intended to demonstrate the practical application of open-access geospatial tools for commercial agricultural monitoring in the West African context.

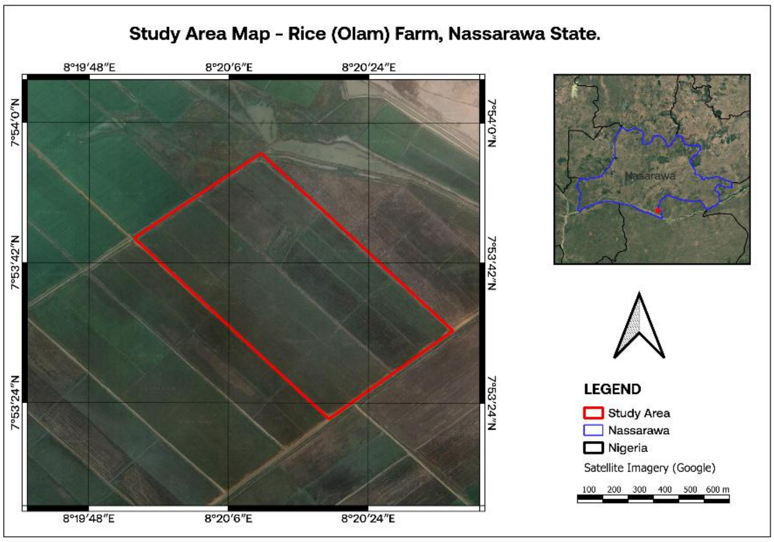
2. STUDY AREA

The study site is a commercial rice farm located in Nassarawa State, North-Central Nigeria, operated by Olam International, one of the largest agribusiness conglomerates operating across Africa. The farm is situated in Akpetche, Nassarawa State, at approximately 7°53'N, 8°20'E, within the Guinea Savanna vegetation zone characterised by a distinct wet season between April and October and a dry season from November to March.

The sample farm covers an area of approximately 61.3 hectares and is cultivated as a single contiguous rice production unit. The field geometry is roughly rectangular, oriented on a northeast-southwest axis, and is surrounded by a mosaic of other agricultural fields and sparse natural vegetation typical of the Guinea Savanna transition zone. A canal system is visible along the northern boundary of the farm, suggesting supplementary irrigation capacity in addition to rain-fed production.

Soil analysis using the iSDAsoil dataset at 30-metre resolution characterised the farm's surface soils (0–20 cm depth) as Sandy Clay Loam, with a sand content of approximately 45.9%, silt content of 25.7%, and clay content of 29.3%. The soil pH was measured at 6.2, which falls within the optimal range for rice cultivation (5.5–7.0). Sandy Clay Loam soils support adequate water retention for paddy rice while providing sufficient drainage between seasons, making them well-suited for commercial rice production.

The study area experiences mean annual rainfall of approximately 1,200–1,500 mm, with peak rainfall occurring between July and September, coinciding with the primary rice growing season. Figure 1 shows the location of the study area within Nassarawa State and Nigeria.



[Figure 1: Study Area Map — Rice (Olam) Farm, Nassarawa State, Nigeria]

3. DATA AND METHODOLOGY

3.1 Satellite Data

Sentinel-2 Level-2A Surface Reflectance imagery was acquired from the Copernicus Sentinel-2 SR Harmonised dataset available on Google Earth Engine. Surface Reflectance data is atmospherically corrected, removing the effects of atmospheric scattering and absorption to provide accurate ground reflectance values suitable for multi-temporal index computation. The Top-of-Atmosphere (TOA) product was deliberately avoided as uncorrected atmospheric effects introduce systematic bias in index values across different acquisition dates and seasons.

Imagery was acquired for three consecutive rice growing seasons, defined as June to November for each of 2023, 2024, and 2025. This window captures the full rice phenological cycle from planting through harvest. A total of 8, 12, and 9 valid cloud-free observations were retained for 2023, 2024, and 2025 respectively after cloud masking and quality filtering. The 2023 season lacked coverage prior to July 18 due to persistent cloud cover during June, and the 2025 season had a significant data gap between June and late September, both of which are acknowledged as limitations inherent to optical remote sensing in tropical regions.

3.2 Cloud Masking

Cloud masking was applied using the Sentinel-2 Scene Classification Layer (SCL), which classifies every pixel in each scene into discrete land cover categories including vegetation, bare soil, water, cloud, cloud shadow, and haze. Pixels classified as vegetation (class 4), bare soil (class 5), and water (class 6) were retained, while all other classes were masked. This pixel-level masking approach was applied after a scene-level pre-filter that excluded scenes with greater than 90% cloud cover metadata. The combination of scene-level pre-filtering and pixel-level SCL masking ensured that only valid ground observations contributed to subsequent analysis. Observations with NDVI values below 0.05 following masking were excluded from time series analysis as these represent residual cloud or shadow contamination rather than valid vegetation signal.

3.3 Vegetation Indices

Two spectral indices were computed for each cloud-masked image in the collection.

The Normalised Difference Vegetation Index (NDVI) was computed using Sentinel-2 Band 8 (Near-Infrared, 842 nm) and Band 4 (Red, 665 nm) according to the formula:

$$\mathbf{NDVI = (B8 - B4) / (B8 + B4)}$$

NDVI values range from -1 to 1, with values above 0.4 generally indicative of healthy green vegetation. For rice crops at peak canopy development, NDVI values typically fall between 0.6 and 0.9. Lower values indicate sparse vegetation, stressed crops, or bare soil conditions.

The Normalised Difference Moisture Index (NDMI) was computed using Sentinel-2 Band 8 (Near-Infrared) and Band 11 (Short-Wave Infrared, 1610 nm) according to the formula:

$$\mathbf{NDMI = (B8 - B11) / (B8 + B11)}$$

NDMI is particularly relevant for rice monitoring as it measures canopy moisture content. High NDMI values indicate water-rich canopy conditions consistent with actively growing paddy rice, while low or negative values indicate moisture stress, senescence, or bare soil. The inclusion of NDMI alongside NDVI provides a more complete characterisation of crop condition than vegetation greenness alone.

3.4 Seasonal Composites and Time Series

Two complementary analytical approaches were applied for each season.

Seasonal median composites were generated by computing the pixel-wise median value across all cloud-free observations within each season window. The median statistic was selected over the mean due to its robustness against outliers, including residual cloud edge pixels and atmospheric anomalies that may persist after masking. Each composite represents the typical vegetation and moisture conditions observed across the full growing season and was used for spatial mapping and statistical extraction.

Full-season time series were generated by extracting the mean NDVI and NDMI value across all farm pixels for each individual cloud-free image date. This produced a temporal profile of index values from June to November for each season, enabling tracking of crop development through planting, tillering, heading, grain filling, senescence, and harvest stages.

All processing was conducted within the Google Earth Engine cloud computing platform. Seasonal median composites were exported as GeoTIFF files at 10-metre spatial resolution in the WGS84 UTM Zone 32N coordinate reference system (EPSG:32632) for cartographic production in QGIS 3.40.

3.5 Statistical Analysis

Descriptive statistics including minimum, maximum, and mean values of NDVI and NDMI were extracted from each seasonal median composite over the farm boundary using the reduceRegion function in Google Earth Engine with a spatial resolution of 10 metres. These statistics provide a quantitative basis for inter-annual comparison of farm performance.

3.6 Soil Data

Soil texture and pH data were obtained from the iSDAsoil dataset, a 30-metre resolution continental African soil property map produced by Innovative Solutions for Decision Agriculture (iSDA) using machine learning trained on over 100,000 analysed soil samples. The dataset is available through Google Earth Engine. Surface layer properties at 0–20 cm depth were extracted for sand content, silt content, clay content, and soil pH over the farm boundary.

4. RESULTS

4.1 Soil Characterisation

Soil analysis using the iSDAsoil dataset characterised the farm's surface soils as Sandy Clay Loam, with mean sand content of 45.9%, silt content of 25.7%, and clay content of 29.3% at 0–20 cm depth. Mean soil pH was 6.2, falling within the optimal range for rice cultivation. The clay fraction of 29.3% supports adequate water retention capacity for paddy rice production, while the dominant sand fraction ensures sufficient drainage between cropping cycles. These soil characteristics are broadly favourable for commercial rice cultivation in the Guinea Savanna agro-ecological zone.

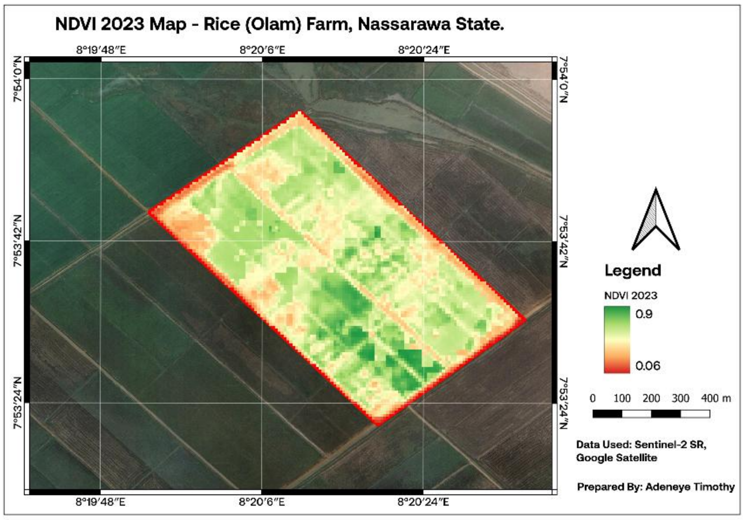
4.2 NDVI Spatial Distribution

Seasonal median NDVI composites revealed distinct spatial patterns of vegetation health across the farm for each of the three growing seasons (Figures 2, 3, and 4).

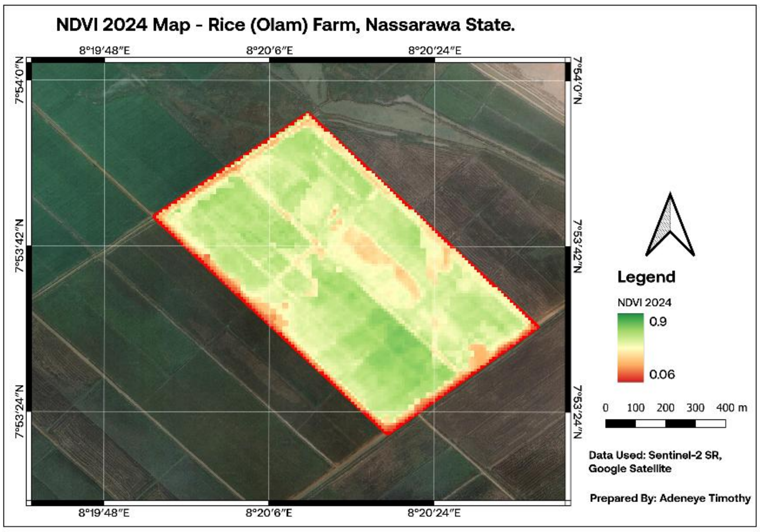
In 2023, the farm exhibited moderate to high NDVI values across much of its extent, consistent with active rice growth during the peak season. The spatial pattern showed a mix of healthy green zones and localised areas of lower NDVI concentrated in the northwestern corner and along internal field access paths, indicating uneven crop establishment or variable soil conditions within that section. The seasonal median composite integrates observations from July through November, meaning the map reflects an averaged view across both peak and declining phases of the 2023 season.

In 2024, the NDVI spatial distribution showed a predominantly green coverage across the farm, with higher overall uniformity relative to 2023. Localised stress zones remained visible in the central section of the farm, appearing as yellow and orange patches. These zones spatially correspond with low NDMI values in the same season, suggesting a persistent area of reduced vegetation density and moisture retention. The extended high-NDVI period observed in the 2024 time series, with values remaining above 0.6 through October and into early November, is reflected in the composite as a generally healthy seasonal signal.

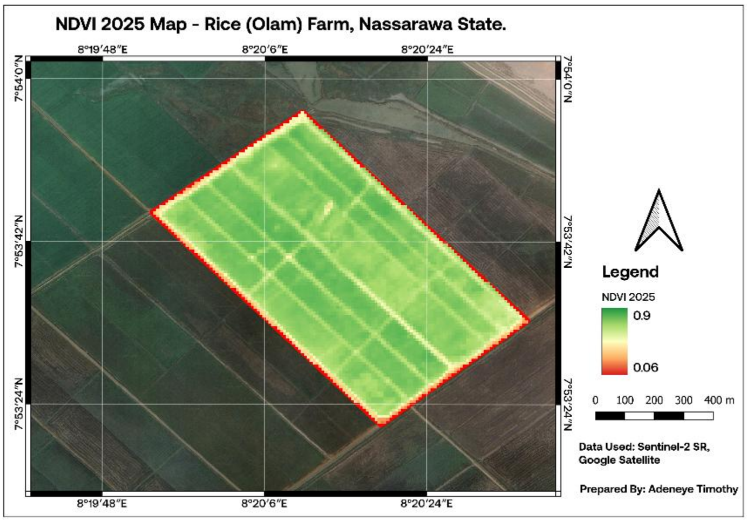
In 2025, the farm displayed the most spatially uniform and extensive NDVI coverage of the three seasons. The composite is dominated by green tones across the farm extent, consistent with the peak NDVI of 0.773 recorded on 4 November 2025. The uniformity of the 2025 spatial pattern suggests more consistent crop establishment and development relative to earlier seasons, despite the significant data gap between June and late September that limits the representativeness of the full-season composite for that year.



[Figure 2: NDVI 2023 Map — Rice (Olam) Farm, Nassarawa State]



[Figure 3: NDVI 2024 Map — Rice (Olam) Farm, Nassarawa State]



[Figure 4: NDVI 2025 Map — Rice (Olam) Farm, Nassarawa State]

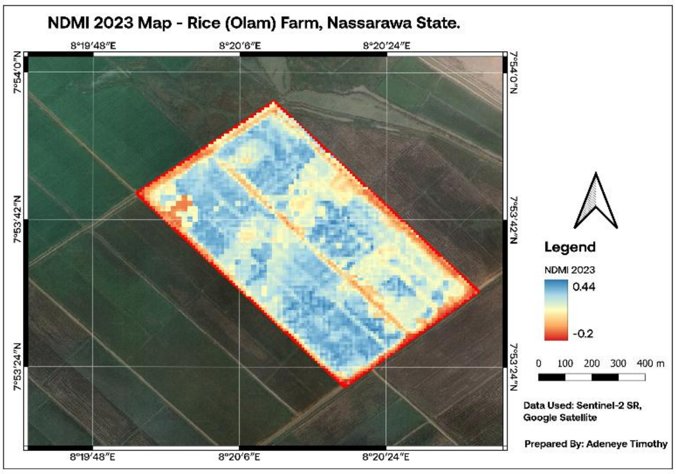
4.3 NDMI Spatial Distribution

Seasonal median NDMI composites revealed the spatial distribution of canopy moisture content across the farm for each season (Figures 5, 6, and 7).

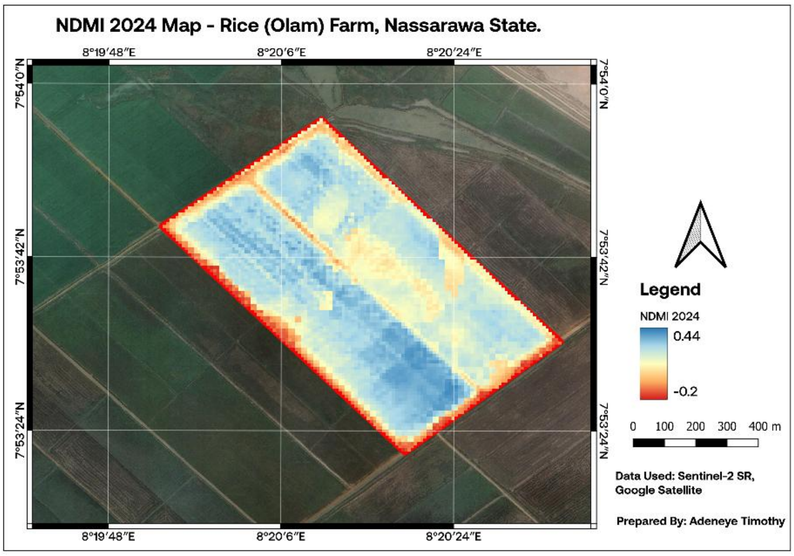
In 2023, the NDMI composite showed a predominantly blue spatial pattern across the farm interior, indicating moderate to high canopy moisture content. Lower moisture zones, represented by yellow and orange tones were concentrated along field boundaries, access paths, and the northwestern section of the farm where NDVI was also comparatively lower. The spatial gradient between high-moisture interior zones and lower-moisture boundary areas is consistent with the paddy structure of the farm, where canopy closure is more complete in the central productive zones.

In 2024, the NDMI spatial pattern showed distinct zonation. The majority of the farm displayed moderate to high moisture values represented by blue tones, while a notable low-moisture zone persisted in the central section, spatially co-located with the low NDVI zone identified in the same season. This dual-index spatial convergence, low NDVI and low NDMI at the same location, strengthens the interpretation of a site-specific constraint affecting both vegetation density and moisture retention in that area.

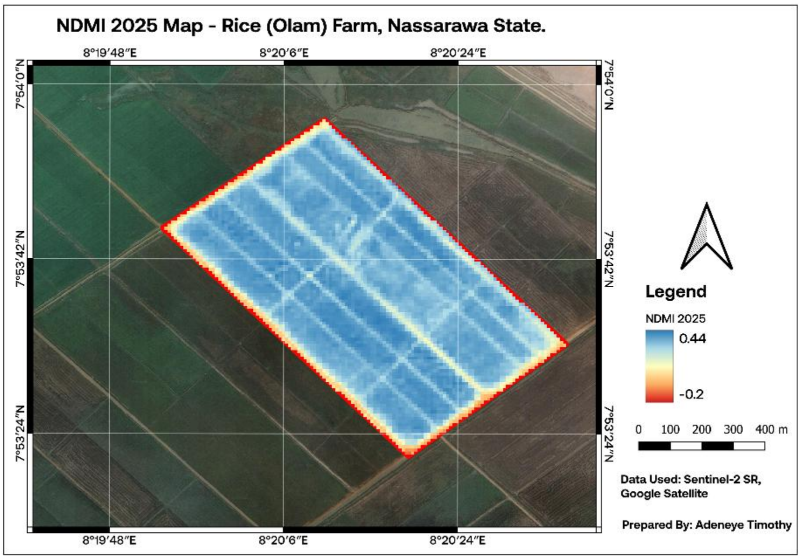
In 2025, the NDMI composite showed the most uniform and extensive high-moisture coverage of the three seasons, consistent with the high and spatially uniform NDVI observed for the same period. The farm displayed predominantly blue tones across its extent, with only minor lower-moisture patches along boundary and access path areas. This pattern is indicative of a well-established, densely canopied rice crop with effective moisture retention at the time of the composite observations.



[Figure 5: NDMI 2023 Map — Rice (Olam) Farm, Nassarawa State]



[Figure 6: NDMI 2024 Map — Rice (Olam) Farm, Nassarawa State]



[Figure 7: NDMI 2025 Map — Rice (Olam) Farm, Nassarawa State]

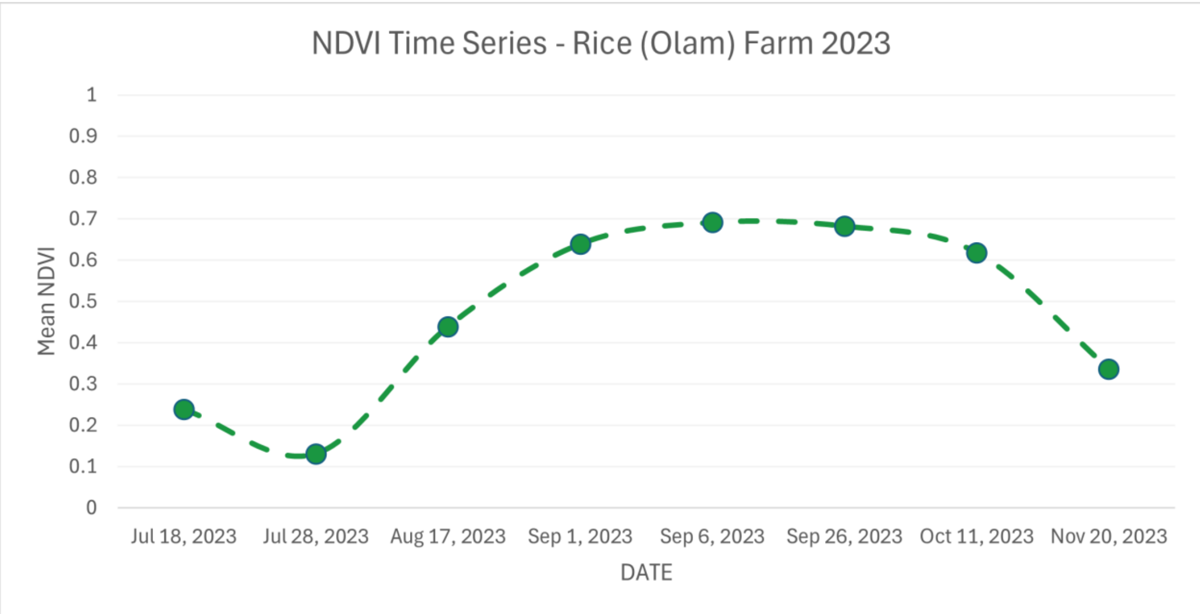
4.4 NDVI Time Series Analysis

Full-season NDVI time series tracked the temporal dynamics of vegetation development across all three growing seasons (Figures 8, 9, and 10).

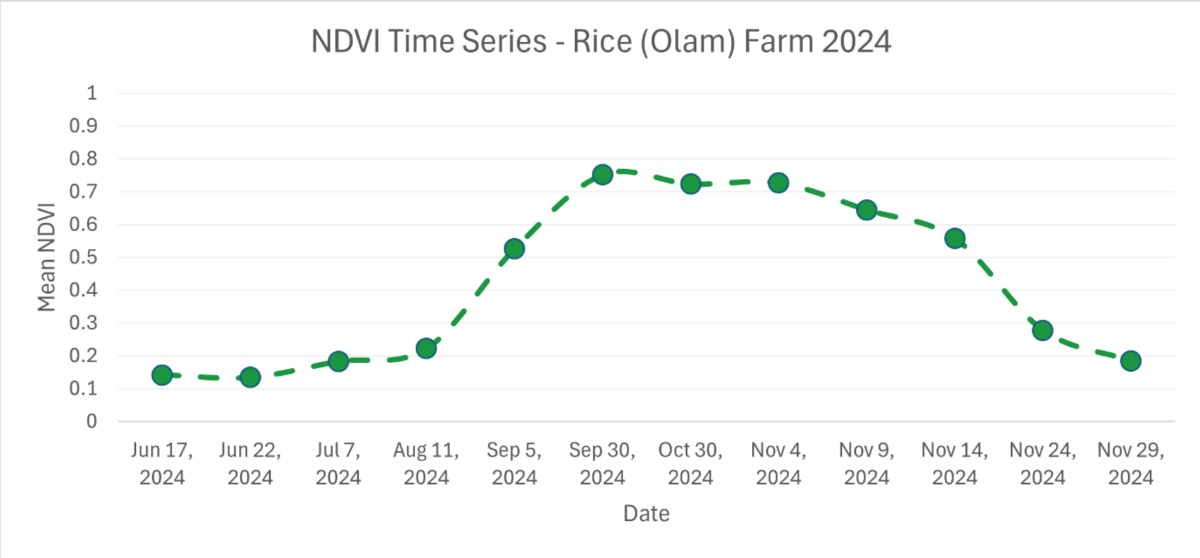
In 2023, the time series began in mid-July with relatively low NDVI values of 0.238 (18 July) and 0.131 (28 July), reflecting early crop establishment following planting. Values rose steadily through August (0.439 on 17 August) and into September, reaching a peak of 0.691 on 6 September, consistent with maximum canopy development at the heading and grain-filling stage. A gradual decline followed through September and October (0.618 on 11 October), with a sharp drop to 0.336 by 20 November, indicating advanced senescence and harvest. The 2023 season shows the earliest and sharpest NDVI peak of the three years, with a clear and well-defined phenological arc despite the absence of June observations.

In 2024, the time series captured a more complete seasonal profile beginning in mid-June with very low values of 0.142 and 0.134, reflecting bare or newly planted conditions. Values remained low through July (0.183) and August (0.222), suggesting a slower canopy development phase relative to 2023. A marked acceleration occurred in September, with NDVI rising from 0.526 on 5 September to 0.752 on 30 September, the highest single-date NDVI value recorded in 2024. Unlike 2023, the high values persisted well into October and November (0.724 on 30 October, 0.727 on 4 November, 0.645 on 9 November), before declining through late November to 0.185 on 29 November. This extended high-NDVI period suggests either a later planting date, a longer-maturing rice variety, or delayed harvest relative to 2023.

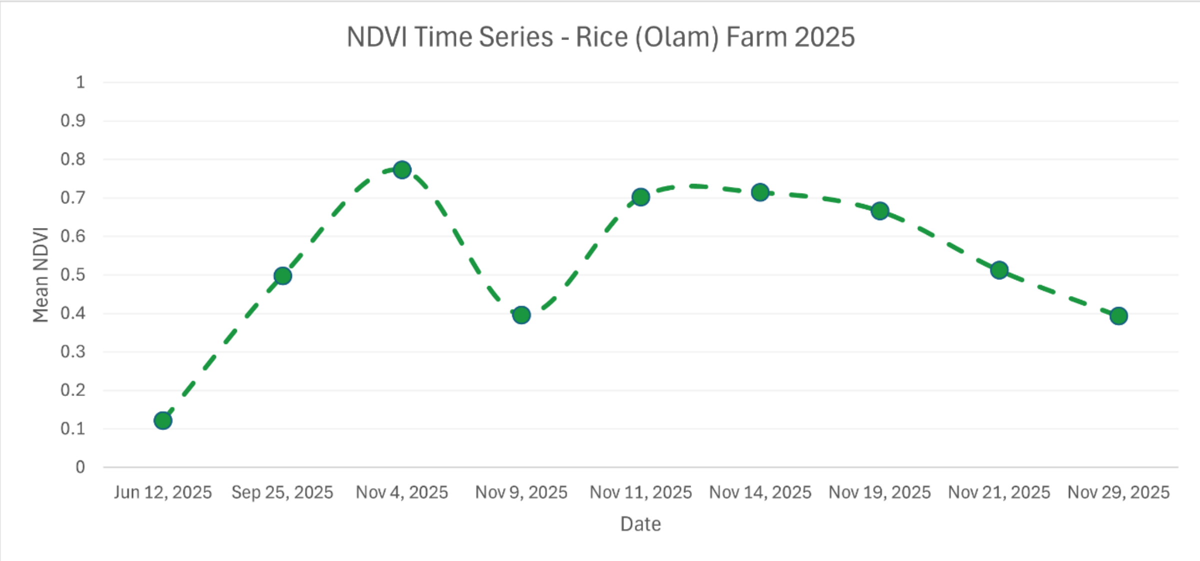
In 2025, the time series was limited by a significant data gap between 12 June (NDVI 0.122) and 25 September (NDVI 0.498), attributable to persistent cloud cover during the July to mid-September period. As a result, the rising phase of the crop cycle, covering planting, tillering, and early heading, is not captured for this season. Available observations show NDVI rising from 0.498 on 25 September to a season peak of 0.773 on 4 November, the highest peak NDVI recorded across all three seasons. Values then declined through November, reaching 0.393 by 29 November. The late-season peak and high maximum NDVI in 2025 suggest strong canopy development, though the absence of mid-season data limits interpretation of the full phenological trajectory.



[Figure 8: NDVI Time Series — Olam Rice Farm 2023]



[Figure 9: NDVI Time Series — Olam Rice Farm 2024]



[Figure 10: NDVI Time Series — Olam Rice Farm 2025]

4.5 NDMI Time Series Analysis

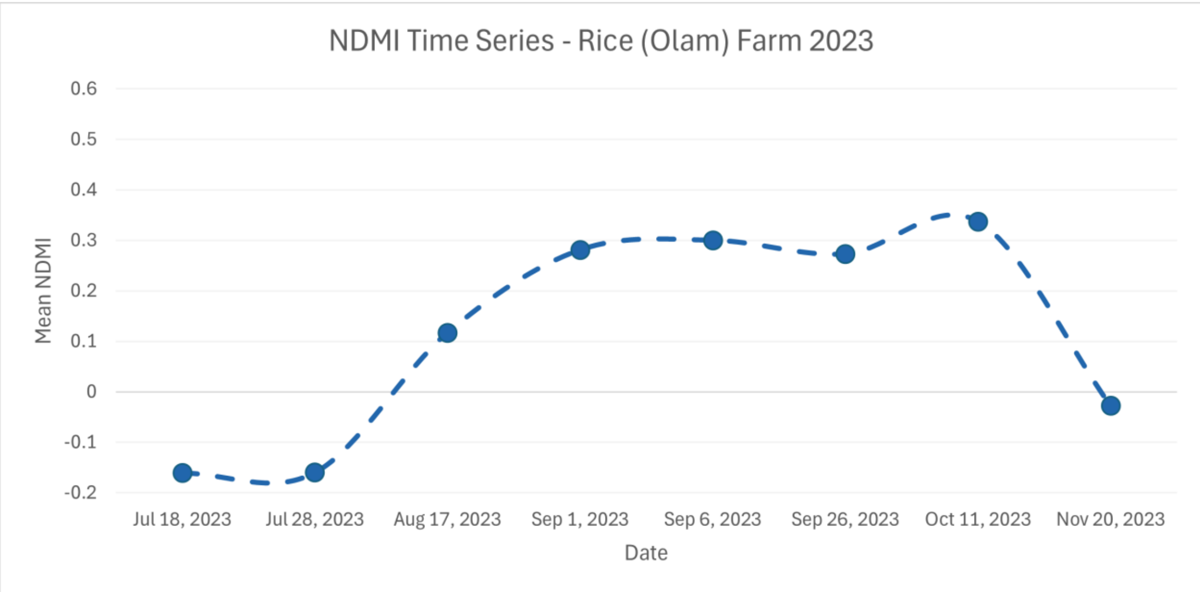
Full-season NDMI time series revealed the temporal dynamics of canopy moisture content across all three growing seasons (Figures 11, 12, and 13).

In 2023, NDMI values began negative in July (-0.161 on 18 July and -0.160 on 28 July), physically consistent with pre-canopy closure conditions when the rice crop had not yet developed sufficient leaf area to retain detectable canopy moisture. Values rose through August (0.117 on 17 August) and September, reaching a peak of 0.337 on 11 October, slightly later than the NDVI peak on 6 September. This lag between NDVI and NDMI peaks is agronomically meaningful; maximum greenness precedes maximum canopy water content as biomass accumulation continues into the grain-filling stage. NDMI dropped sharply to -0.028

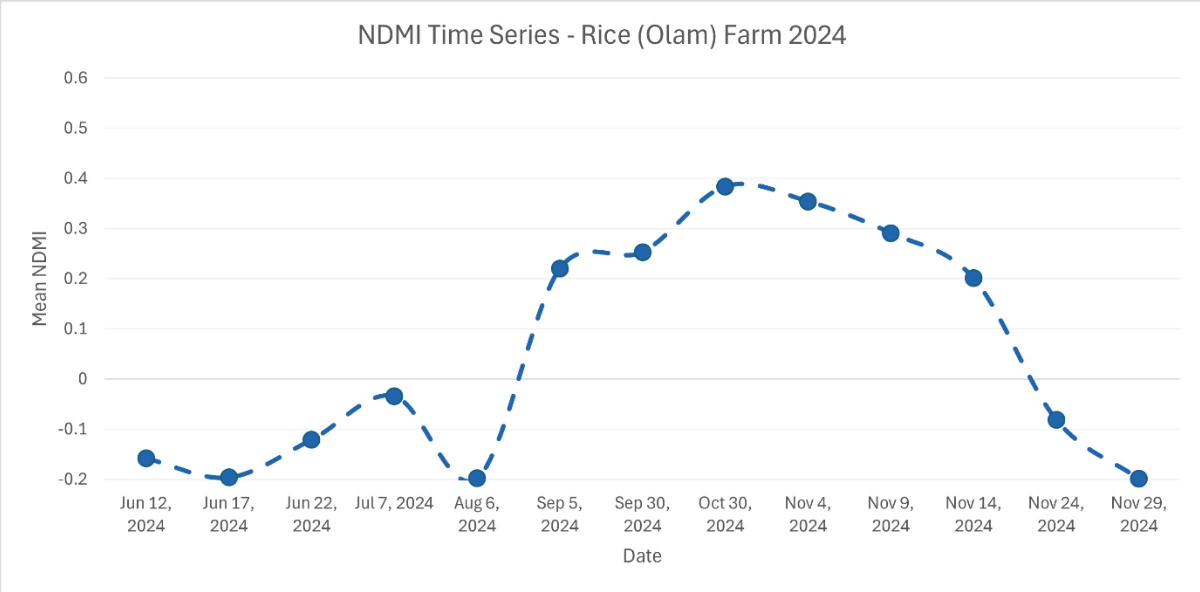
by 20 November, confirming crop senescence and harvest consistent with the NDVI decline observed at the same date.

In 2024, the NDMI time series showed a more complex pattern. Values were negative throughout June and July (-0.158 to -0.034), consistent with sparse early-season canopy. An anomalous low value of -0.198 on 6 August is likely attributable to residual cloud or atmospheric contamination that passed the initial quality filters, and should be interpreted with caution. From September onwards, NDMI rose consistently from 0.220 on 5 September to a peak of 0.384 on 30 October, mirroring the extended high-NDVI period observed in the same season. Values declined sharply in late November, reaching -0.081 and -0.199 on 24 and 29 November respectively, indicating rapid canopy collapse consistent with harvest.

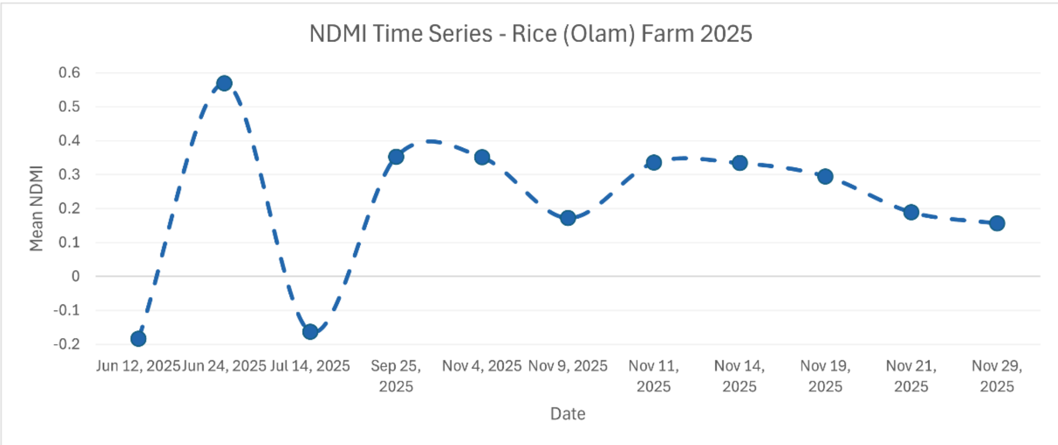
In 2025, the NDMI time series was also affected by the mid-season data gap. An early low value of -0.183 on 12 June reflects bare soil conditions at season onset. A notably high value of 0.569 on 24 June is anomalous relative to the surrounding observations and is likely a cloud-contaminated outlier rather than a genuine vegetation signal; it should be treated as unreliable. The subsequent value of -0.163 on 14 July confirms that canopy moisture was still low at that point. From late September, NDMI values rose to 0.353 on 25 September and remained moderately elevated through November, declining from 0.351 on 4 November to 0.156 on 29 November. The 2025 NDMI peak was lower than 2024 (0.353 vs 0.384), despite NDVI being higher, which may reflect differences in canopy structure or water use efficiency between seasons.



[Figure 11: NDMI Time Series — Olam Rice Farm 2023]



[Figure 12: NDMI Time Series — Olam Rice Farm 2024]



[Figure 13: NDMI Time Series — Olam Rice Farm 2025]

4.6 Summary Statistics

Table 1 presents the descriptive statistics of NDVI and NDMI extracted from seasonal median composite rasters over the farm boundary for each growing season. Minimum and maximum values reflect the spatial range of index values across all farm pixels in the composite, while mean values represent the average condition across the farm extent.

Season	NDVI Min	NDVI Max	NDVI Mean	NDMI Min	NDMI Max	NDMI Mean
2023	0.064	0.901	0.554	-0.222	0.423	0.213
2024	0.126	0.760	0.559	-0.182	0.413	0.208
2025	0.191	0.812	0.706	-0.112	0.435	0.330

Table 1: Summary Statistics of NDVI and NDMI — Olam Rice Farm 2023–2025

5. DISCUSSION

5.1 Inter-Annual Trends in Vegetation Health

The three-season NDVI analysis reveals a progressive increase in peak canopy greenness from 2023 to 2025, with maximum NDVI values of 0.691, 0.752, and 0.773 recorded respectively. This upward trend in peak NDVI is consistent with improving crop performance over the study period, though the interpretation must account for inter-annual differences in season timing and data availability that complicate direct comparison.

The 2023 season showed the earliest and most clearly defined phenological arc, with NDVI peaking in early September and declining sharply by November. This pattern is consistent with a timely planting and well-managed growing season, though the absence of June observations means the planting phase is not directly observed. The spatial NDVI map for 2023 shows moderate heterogeneity, with stressed zones in the northwestern section suggesting localised management or soil variability issues.

The 2024 season was characterised by a slow early-season rise and a late, sustained peak extending through October and into November. This extended high-NDVI period could reflect a later planting date, a longer-duration rice variety, or delayed harvest. The spatial map confirms generally healthy canopy coverage but with a persistent central stress zone that co-occurs with low NDMI, suggesting a site-specific constraint requiring investigation.

The 2025 season produced the highest peak NDVI of the three years (0.773 on 4 November), but the significant data gap between June and late September means the ascending phase of the crop cycle is unobserved. The spatial map and available late-season observations confirm strong canopy development, but conclusions about early-season conditions in 2025 cannot be drawn from the available data.

5.2 Canopy Moisture Dynamics and Implications for Rice Production

The NDMI time series across all three seasons consistently showed negative values at season onset, transitioning to positive values as the rice canopy developed. This pattern is physically meaningful and validates the index's sensitivity to paddy rice phenology: negative NDMI reflects the low canopy moisture of bare or sparsely vegetated soil at planting, while positive values confirm active moisture retention as the leaf area index increases through tillering and heading.

A consistent feature across all three seasons is the lag between the NDVI peak and the NDMI peak. In 2023, NDVI peaked on 6 September while NDMI peaked on 11 October. In 2024, NDVI peaked on 30 September while NDMI peaked on 30 October. This one-to-four-week lag is agronomically consistent with the rice growth cycle, where maximum greenness occurs at heading while maximum canopy water content continues to build through the grain-filling stage as biomass accumulates.

The close spatial correspondence between low NDVI and low NDMI zones in 2024 strengthens confidence in the identification of a persistent stress area in the central section of the farm. When two independent spectral indices derived from different parts of the electromagnetic spectrum converge on the same spatial anomaly, the probability of a genuine ground condition rather than a sensor or processing artefact is substantially higher. This convergence of evidence is a key methodological strength of the dual-index approach.

Two data quality anomalies warrant explicit acknowledgement. The NDMI value of -0.198 recorded on 6 August 2024 and the value of 0.569 recorded on 24 June 2025 are inconsistent with surrounding observations and likely represent residual cloud or atmospheric contamination that passed the quality filters. These points were retained in the time series charts for transparency but should not be interpreted as genuine vegetation signals.

5.3 Methodological Considerations

Several methodological considerations are relevant to the interpretation of these findings. The 2023 season lacked satellite coverage prior to mid-July, and the 2025 season had a significant gap between mid-June and late September. These gaps are inherent to optical remote sensing in humid tropical regions where persistent cloud cover during the early wet season limits data availability. They do not affect the validity of the peak-season spatial analysis but limit the completeness of the time series for those years.

The use of seasonal median composites integrates observations across the full June to November window. This maximises spatial coverage but means the composite does not represent a single phenological moment. Pixels from different dates contribute depending on cloud patterns, introducing temporal mixing. This is standard practice in operational agricultural remote sensing and is acknowledged as a characteristic of the methodology rather than a flaw.

The 10-metre spatial resolution of Sentinel-2 means that internal farm features; access paths, bunds, irrigation channels, occupy multiple pixels and contribute to the low index values observed along linear features within the farm boundary. These are not indicative of crop stress but of non-crop surfaces.

The absence of ground-truth data including yield records, planting dates, fertilisation schedules, and irrigation logs means that the drivers of observed inter-annual trends cannot be definitively attributed to specific factors. The satellite evidence is consistent with generally improving and variable performance but cannot alone distinguish between management-driven and climate driven variation. Integration of field records with satellite analysis would substantially strengthen interpretive power.

6. CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

This study demonstrated the capability of freely available Sentinel-2 satellite imagery, processed through Google Earth Engine, to monitor vegetation health and canopy moisture dynamics at a commercial rice farm in Nassarawa State, Nigeria over three consecutive growing seasons. The multi-temporal NDVI and NDMI analysis produced both spatial maps and temporal profiles that together characterise farm performance at a level of detail and consistency that would be impractical to achieve through conventional field-based monitoring alone.

Peak NDVI values showed a progressive increase across the three seasons; 0.691 in 2023, 0.752 in 2024, and 0.773 in 2025, consistent with improving canopy development at the height of each growing season. NDMI time series confirmed the expected seasonal moisture trajectory in all three years, with negative values at transplanting transitioning to positive peaks at canopy closure, and a consistent lag between the NDVI and NDMI peaks that reflects the progression from maximum greenness through grain filling. Spatially, both indices identified a persistent stress zone in the central section of the farm in 2024, where co-located low NDVI and NDMI values indicate a site-specific constraint warranting ground investigation.

Data availability limitations, particularly the absence of mid-season observations for 2023 and 2025 due to cloud cover, constrain the completeness of the phenological record for those years. These limitations are inherent to optical remote sensing in tropical West Africa and underscore the value of complementary radar-based monitoring using Sentinel-1 SAR data, which is not affected by cloud cover.

Taken together, the findings demonstrate that open-access satellite data and cloud computing platforms can deliver actionable agricultural intelligence for commercial rice operations in Nigeria without requiring expensive proprietary data or ground-based sensor networks. This has significant implications for the scalability of farm performance monitoring across the broader smallholder and commercial rice production landscape in Nigeria.

6.2 Recommendations

Based on the findings of this study, the following recommendations are made for farm managers, agricultural technology practitioners, and future research:

Investigate the persistent central stress zone identified in 2024. The spatial co-occurrence of low NDVI and low NDMI in the central section of the farm in 2024 warrants ground-level investigation. Soil sampling, compaction testing, and drainage assessment in this zone would help confirm whether the anomaly reflects a structural soil issue, irrigation distribution inefficiency, or access path compaction, and guide targeted remediation before it reoccurs in future seasons.

Integrate Sentinel-1 SAR data to address cloud cover limitations. The data gaps in 2023 and 2025 due to persistent cloud cover represent a significant constraint on the completeness of optical time series analysis in this region. Sentinel-1 Synthetic Aperture Radar data, which penetrates cloud cover, should be incorporated alongside Sentinel-2 to provide continuous monitoring through the early wet season when planting and early canopy development occur.

Extend the monitoring framework to include additional indices. The addition of the Normalised Difference Water Index (NDWI) would provide explicit monitoring of open water surface dynamics within the paddy, complementing the canopy-level moisture information provided by NDMI. The inclusion of a red-edge based index such as the Chlorophyll Red-Edge Index (CIre) would improve sensitivity to early nitrogen stress, which is not always detectable in broadband NDVI until stress is advanced.

Integrate field records with satellite analysis. The interpretive power of satellite-based monitoring is substantially enhanced when combined with field management records including planting dates, fertilisation schedules, irrigation logs, and yield data. Establishing a systematic data collection protocol alongside the satellite monitoring framework would enable causal attribution of observed trends and support predictive modelling of yield outcomes from satellite inputs.

Scale the methodology to additional farms. The workflow developed in this study, from Google Earth Engine processing through QGIS cartographic production, is fully replicable and scalable. Applying the same framework across multiple farms within a portfolio would enable comparative performance benchmarking, identification of best-performing management practices, and early warning of emerging stress conditions at landscape scale. This is directly relevant to agricultural technology platforms operating across smallholder and commercial farm networks in Nigeria.

Transition to near-real-time monitoring. The current study used seasonal median composites for spatial mapping. A near-real-time implementation using individual Sentinel-2 acquisitions processed automatically through Google Earth Engine would enable in-season crop monitoring and timely intervention recommendations rather than retrospective analysis. This represents the logical next step toward an operational farm intelligence system.

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APPENDIX – GEE SCRIPT

The complete Google Earth Engine JavaScript code used for data acquisition, cloud masking, index computation, time series extraction, and data export is available at the project GitHub repository.

GitHub: https://github.com/Bloom9ja/